

3 Miscon. AI will replace humans on every job.

Reality AI automates repetitive tasks but still need humans for creativity, emotions and decision-making.

ex: AI can help doctors analyze medical scans, but humans are still need for diagnosis and patient care.

⇒ Assumptions behind AI research.

AI research is based on certain assumptions that guide how scientists and engs develop AI systems.

1 Intelligence can Be represented as Data.

Assumption: Human intelligence can be understood and modeled using data and Algorithms.

→ Researchers believe that if we collect enough data and develop the right algo we can replicate or simulate intelligence in machines.

2 AI can improve through learning.

Assumption: Machine can learn from data and improve over time without needing explicit prog for every task.

3 Problem have optimal solutions.

Cognitive science:

⇒ Distinctions Rational vs human.

→ AI researchers compare two types of thinking:

1. Rational thinking: Thinking logically and always making the best decision.

2. Human thinking: Thinking like a person, which includes emotions, biases and mistakes.

• AI's Goals: Rational or Human?
Old AI focused on mimicking human behaviour
eg: chatbots.

• Modern AI focuses on rational thinking
(eg recommendation systems, self driving cars)

⇒ Common Misconceptions:

1. Misconception: AI has emotions, creativity, and personal thoughts.

Reality: AI follows math and logic, it doesn't feel or think like humans.

2. Misconception: AI keeps learning by itself like a human.

Reality: AI needs training data and human supervision to improve.

ex: A self driving car needs millions of labelled images before it can recognize traffic signs, it doesn't just figure it out like a human child would.

⇒ characterizing AI Systems: AI systems can be characterized based on how they think, learn, and act. Here are the main ways to describe them.

① Based on learning Types

↳ Supervised learning: learns from labeled data

ex: An AI trained with images of cats and dogs, where each image is labeled "cat" or "dog"

② Unsupervised learning: finds patterns without labeled data.

ex: AI grouping customers based on shopping habits without knowing their preferences beforehand.

③ Reinforcement learning: learns by trial and error, like a game.

ex:

A robot that tries different paths in a maze and learns the best way.

→ Think vs act:

→ Think is the internal process which involves mental activity: processing information, forming ideas, reasoning, planning, and completing.

Abstract: Think can occur without any outward manifestation, you can think about a problem for hours without taking physical action.

→ Acting:

External manifestation.

This involves physical or observable behavior, carrying out a task, making a decision, or interacting with the world.

• it is the translation of thoughts and intentions into concrete action.

2/ Theory of mind intelligence
Understand emotions and thoughts (not fully developed yet) ex: A future robot that understands human feelings

3/ Limited memory intelligence:-
Remembers past data to improve
ex: self-driving cars that learn from past experience.

→ Simple: AI aims to create machine that can "think" and "learn" like humans, allowing them to handle complex tasks.

⇒ characteristics of AI:

1. AI can get better at tasks over time by looking at data.

2. Problem-solving.

AI can figure out solutions to complex issues.

3. Reasoning. - AI can use logic to make decisions.

4. Automation.

AI can do tasks automatically without needing constant human help.

→ What do people in AI think about Intelligence?

In AI Intelligence means the ability to learn, reason, and make decisions like a human but using computers. AI researchers try to teach machines to think and act smartly based on data and experiences.

⇒ Types of intelligence in AI:

1. Reactive intelligence:- only responds to situations, no learning.

ex: A chess-playing AI that only follows programmed rules.

→ AI: The Birth of ai (1950's)

→ Early concepts of AI emerged in the 1940's and 1950's.

→ Alan Turing's 1936 paper laid the theoretical foundation for AI and introduced the concepts of a universal machine capable of performing any computation.

→ ^{In} After 1950 some computerest concept were keep that some computer technologies could be occur which will think like as a human.
means why chawna parya computer jo software thi saghan tha oon.

→ John McCarthy give the name of AI. Intelligence like human have ability to think it self and take decisions.

so Artificial Intelligence means:-

Introd to Ai:

1. Making computer smarter it's teaching computer to do things that usually require human intelligence.

② Thinking and Learning:- This includes understanding, Recognizing, solving problems, learning from data.

③ Automating tasks:- AI helps computers perform tasks without needing constant human input.

⇒ Application of python

- 1, Website and application development (Django framework)
- 2, Desktop app (Tkinter, PyQt)
- 3, Game Develop
- 4, Web scraping
- 5, Data visualization.
- 6, Scientific calc (Machine learn)
- 7, Machine learn and AI.
- 8, 3D App Dev.

• Google, Netflix, Facebook, Youtube use python.

→ Note: python has one Dependency Manager.
if we want to install any package, then we
can install ^{through} pip.

so while installing python pip should be
taken

→ There are so many IDEs one of them
is Pycharm

⇒ Python is an an interpreted, Object oriented, Name = Noor Mustafa
high-level programming languages with copy: Python
Dynamic semantics.

→ interpreter means How it is different from java, c++,
java and c++ work compiler base.

interpreter means if i have write the code over python
then example we have write 3 line of code.
and interpreter will check the code line by line.

Note In Interpreter if there is error in second line then
it will give you the output of first line.

and in compiler until you will not ~~done~~ your whole
program it will not run/execute.

Compiler execute the whole code at once.

it is also object oriented.

Highlevel means, which is human understandable like a/clo.

dynamic semantics means, it will identify in
run time itself that a/clo. that here a/clo is
integer.

→ python syntax are easy compared to other program
languages.
ex print(a+b)

⇒ History of Python it was introduce by Guido van
Rossum in 1989.

First version was released in 1991

in python 2.x also add some features

like comprehensions, garbage collection system.

After python 2.x python 3.x released in 2008.